Applied Algebra

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Last updated: March 12, 2026

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1 The Binomial Theorem

Theorem 1 (Binomial Theorem)**.** *The Binomial theorem describes the algebraic expansion of powers of a binomial.*

$$(x+y)^n = \binom{n}{0}x^n y^0 + \binom{n}{1}x^{n-1}y^1 + \ldots + \binom{n}{n}x^0 y^n. \tag{1}$$

2 Problem set.

2.1 Algebra 101.

2.1.1 Problem 1.1.

Expand $(a \pm b)^5$.

$$(a \pm b)^5 \tag{2}$$

Using the Binomial Theorem (Theorem 1) yields:

$$(a \pm b)^5 = \binom{5}{0}a^5 \pm \binom{5}{1}a^4 b + \binom{5}{2}a^3 b^2 \pm \binom{5}{3}a^2 b^3 + \binom{5}{4}ab^4 \pm \binom{5}{5}b^5. \tag{3}$$

2.1.2 Problem 1.2.

Expand $(x \pm 1)^5$.

$$(x \pm 1)^5 \tag{4}$$

Using the Binomial Theorem (Theorem 1) yields:

$$(x \pm 1)^5 = \binom{5}{0}x^5 \pm \binom{5}{1}x^4 + \binom{5}{2}x^3 \pm \binom{5}{3}x^2 + \binom{5}{4}x \pm \binom{5}{5}. \tag{5}$$

2.1.3 Problem 1.3.

Expand $(a+b+c+d)^2$.

$$(a+b+c+d)^2 = a^2 + b^2 + c^2 + d^2 + 2ab + 2ac + 2ad + 2bc + 2bd + 2cd. \tag{6}$$

2.1.4 Problem 1.4.

Rewrite $a^2 + ab + a + b$ as a product of two first-degree polynomials.

$$a^2 + ab + a + b = a(a+b) + (a+b) = (a+b)(a+1). \tag{7}$$

2.1.5 Problem 1.5.

Rewrite x^3+2x^2+2x+1 as a product of a second and a first degree polynomial. (It cannot be rewritten as a product of three first-degree polynomials, due to The Fundamental Theorem of Algebra and results from Abstract Algebra / Field Theory).

$$x^3+2x^2+2x+1 = (x^3+x^2+x)+(x^2+x+1) = x(x^2+x+1)+(x^2+x+1) = \tag{8}$$

$$= (x^2+x+1)(x+1). \tag{9}$$

2.1.6 Problem 1.6.

Rewrite $x^3+6x^2+12x+8$ as a product of polynomials.
Method 1.
Notice that the coefficients 1, 6, 12, 8 match the binomial expansion pattern of a perfect cube. Then use the identity

$$a^3+3a^2b+3ab^2+b^3 = (a+b)^3. \tag{10}$$

Thus,

$$x^3+6x^2+12x+8 = (x+2)^3. \tag{11}$$

Method 2.
We observe that -2 is a root of our polynomial (i.e. a point at which the expression evaluates to 0), and thus $(x+2)$ is a factor. Dividing the polynomial by $(x+2)$ gives

$$x^2+4x+4 = (x+2)^2, \tag{12}$$

Therefore

$$x^3+6x^2+12x+8 = (x+2)(x+2)^2 = (x+2)^3. \tag{13}$$

2.1.7 Problem 1.7.

Find all solutions of the equation $ab+2a-3b=14$ where $a, b \in \mathbb{N}^+$.
Group terms containing a

$$ab+2a-3b=14$$

$$a(b+2)-3b=14$$

Move the $3b$ term

$$a(b+2)=3b+14$$

Express a:

$$a=\frac{3b+14}{b+2}$$

Rewrite the numerator:

$$3b+14=3(b+2)+8$$

Thus

$$a = 3 + \frac{8}{b+2}.$$

For a to be integer, $b+2$ must divide 8.
Divisors of 8: $1, 2, 4, 8$. Since $b \geq 1$, $b+2 \geq 3$, we are left with two possibilities:

- $b+2 = 8 \implies b = 2$
- $b+2 = 8 \implies b = 6$

Now we plug these b values back into our expression for a to find the two solutions to our equation: $(5, 2)$ and $(4, 6)$.

2.1.8 Problem 1.8.

What is the minimal point of the expression $2x^2 + 2y^2 - 2xy + 4x - 6y + 8$?

2.1.9 Problem 1.9.

Prove the following inequality:

$$(a+b+c)(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}) \geq 9$$

where $a, b, c > 0$.

2.1.10 Problem 2.1.

Expand $(2x-3)^3$.

2.1.11 Problem 2.2.

Expand $(a-1)(b-1)(c-1)$.

2.1.12 Problem 2.3.

Rewrite the following expression as a product:

$$(a+b)^2 - (a-b)^2. \tag{14}$$

2.1.13 Problem 2.4.

Rewrite the following expression as a product:

$$x^2y + xy^2 - x^2 + y^2. \tag{15}$$

2.1.14 Problem 2.5.

Rewrite $a^4 - 1$ as a product.
Using difference of squares yields:

2.1.15 Problem 2.6.

Find all solutions of the equation

$$4ab - 6a + 8b = 14. \tag{16}$$

2.1.16 Problem 2.7.

Find the minimal point of the following expression

$$5x^2 + 2y^2 - 2x + 4y + 4xy + 1.$$

2.1.17 Problem 2.8.

Prove the following inequality:

$$4x + \frac{1}{x} \geq 4$$

where $x > 0$.

3 Sum of numbers, sum of squares, sum of cubes...

$$\begin{aligned}
1^2 &= 1^2 && = 1^2 \\
2^2 &= (1+1)^2 = 1^2 + 2 \cdot 1 \cdot 1 + 1^2 && = 1^2 + 2 \cdot 1 + 1 \\
3^2 &= (2+1)^2 = 2^2 + 2 \cdot 2 \cdot 1 + 1^2 && = 2^2 + 2 \cdot 2 + 1 \\
4^2 &= (3+1)^2 = 3^2 + 2 \cdot 3 \cdot 1 + 1^2 && = 3^2 + 2 \cdot 3 + 1
\end{aligned}$$

...

$$n^2 = ((n-1)+1)^2 = (n-1)^2 + 2 \cdot (n-1) \cdot 1 + 1^2 = (n-1)^2 + 2(n-1) + 1$$

$$(n+1)^2 = (n+1)^2 = n^2 + 2 \cdot n \cdot 1 + 1^2 = n^2 + 2 \cdot n + 1$$

$$1^2 + 2^2 + ... + n^2 + (n+1)^2 = 1^2 + 2^2 + ... + n^2 + 2 \cdot (1 + 2 + ... + n) + (n+1)$$

$$\begin{aligned}
(n+1)^2 &= 2 \cdot (1 + 2 + ... + n) + (n+1) \\
n^2 + 2n + 1 - n - 1 &= 2(1 + 2 + ... + n) \\
\frac{n(n+1)}{2} = \frac{n^2+n}{2} &= 1 + 2 + ... + n
\end{aligned}$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$\begin{aligned}
1^3 &= 1^3 && = 1 \\
2^3 &= (1+1)^3 = 1^3 + 3 \cdot 1^2 \cdot 1 + 3 \cdot 1 \cdot 1^2 + 1^3 && = 1^3 + 3 \cdot 1^2 + 3 \cdot 1 + 1 \\
3^3 &= (2+1)^3 = 2^3 + 3 \cdot 2^2 \cdot 1 + 3 \cdot 2 \cdot 1^2 + 1^3 && = 2^3 + 3 \cdot 2^2 + 3 \cdot 2 + 1
\end{aligned}$$

$$4^3 = (3+1)^3 = 3^3 + 3\cdot 3^2 \cdot 1 + 3\cdot 3\cdot 1^2 + 1^3 = 3^3 + 3\cdot 3^2 + 3\cdot 3 + 1$$
$$5^3 = (4+1)^3 = 4^3 + 3\cdot 4^2 \cdot 1 + 3\cdot 4\cdot 1^2 + 1^3 = 4^3 + 3\cdot 4^2 + 3\cdot 4 + 1$$
$$(n+1)^3 = \qquad\qquad = n^3 + 3\cdot n^2 + 3\cdot n + 1$$

$$1^3+2^3+...+n^3+(n+1)^3 = 1^3+2^3+...+n^3+3\cdot(1^2+2^2+...+n^2)+3\cdot(1+2+...+n)+(n+1)$$

$$(n+1)^3 = 3\cdot(1^2+2^2+...+n^2) + 3\cdot\frac{n(n+1)}{2} + (n+1) \qquad /\cdot 2$$

Let $1^2 + 2^2 + ... + n^2 = x$

$$2\cdot(n+1)^3 = 6x + 3n(n+1) + 2(n+1)$$

$$2(n+1)^3 - 3n(n+1) - 2(n+1) = 6x$$
$$(n+1)(2\cdot(n+1)^2 - 3n - 2) = 6x$$
$$(n+1)(2n^2 + 4n + 2 - 3n - 2) = 6x$$

$$(n+1)(2n^2+n) = 6x$$
$$6x = n(n+1)(2n+1)$$
$$1^2+2^2+3^2+...+n^2 = \frac{n(n+1)(2n+2)}{6}$$